



SIKSHA 'O' ANUSANDHAN

(Deemed to be University)

Faculty of Engineering & Technology (ITER)

Department of Computer Science and Engineering

Project Proposal Form

FINAL YEAR RESEARCH PROJECT-2026

SECTION: 2241028

GROUP NO: 011

PROJECT TITLE: Breast Cancer Detection Using Machine Learning

PROJECT ABSTRACT:

Breast cancer remains one of the most serious global health challenges and is the most common cancer among women worldwide. It is also a leading cause of cancer-related mortality. Early detection significantly improves survival rates and treatment outcomes. In recent years, Machine Learning (ML) techniques have shown remarkable success in medical diagnosis by analyzing complex patterns in clinical data. This project presents a machine learning-based breast cancer detection system using the Wisconsin Breast Cancer (Diagnostic) dataset available in Python's sklearn datasets. The dataset contains 569 samples with 30 numerical features extracted from Fine Needle Aspiration (FNA) images of breast masses. The objective of this project is to develop an accurate classification model that can distinguish between benign and malignant tumors.

Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel is implemented as the primary classification algorithm due to its strong performance in medical diagnostic tasks. Data preprocessing techniques such as feature scaling using StandardScaler and cross-validation are applied to enhance model performance. Hyperparameter tuning (C and gamma) is performed using Grid Search to maximize classification accuracy. The model's performance is evaluated using metrics such as accuracy, precision, recall, F1-score, and ROC- AUC. The system aims to achieve good performance for breast cancer detection. Additionally, the SVM model is compared with other classifiers such as Logistic Regression and Random Forest to validate its effectiveness. The developed system functions as a decision-support tool that can assist clinicians in early breast cancer diagnosis. By automating tumor classification, the system helps reduce diagnostic errors, improve screening efficiency, and support timely treatment decisions, ultimately contributing to better patient outcomes.

(1) **SOFTWARE, HARDWARE OR METHODS/ALGORITHMS SPECIFICATIONS:**

Software Specification:

Programming Language: Python 3.x

Libraries: Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn

Development Environment: Jupyter Notebook / VS Code

Operating System: Windows 10/11

Processor: Intel Core i5 or equivalent

RAM: Minimum 8 GB

Storage: 20 GB free space

System Type: 64-bit OS

Primary Algorithm: Support Vector Machine (SVM) with RBF kernel

Hyperparameter Tuning: Grid Search (C and Gamma)

Preprocessing: Feature scaling (StandardScaler), Train-Test Split (80:20), Cross-Validation

Evaluation Metrics: Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix

Comparison Models: Logistic Regression, Random Forest

(2) **NAME, REG. NO AND SIGNATURE OF GROUP MEMBERS:**

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(3) **APPROVAL STATUS (To be filled in by the Section Coordinator of FRP):**

Subhashree Mohapatra

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Project Supervisor

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